



FROM YOU

YOUR SUGGESTIONS AND THINGS YOU WANTED TO KNOW!

IDEAS TO DISCUSS

I have a few ideas that I want to discuss. Is there any portal of EFY where I can share my ideas and discuss with experts?

Kumar

EFY. You can discuss your ideas with experts on EFY forum <https://www.electronicsforu.com/forums>

• • •

UPCOMING EVENTS

Can EFY let us know about upcoming events related to electronics so that we can know in advance to register for attending them?

Naveen

EFY. Calendar of events is published under Industry News in the magazine. You can also see such information at <https://www.electronicsforu.com/events>

• • •

DESIGNING AMPLIFIERS

I have seen an amplifier circuit in EFY magazine but I want to customise the circuit according to my need. Since it takes a lot of time to calculate the gain of each op-amp and make a prototype, is there any tool that can help me expedite such designs?

Vishal

EFY. You can quickly do such calculation for your design using our calculator available at <https://www.electronicsforu.com/special/inverting-op-amp-voltage-gain-calculator>

• • •

VARIABLE POWER SUPPLY

This is regarding the excellent DIY article 'Variable Power Supply For Up To 3A Current' published in April issue (also available at [https://www.electronicsforu.com/electronics-pro-](https://www.electronicsforu.com/electronics-pro-jects/variable-power-supply-for-up-to-3a-current)

[jects/variable-power-supply-for-up-to-3a-current](https://www.electronicsforu.com/electronics-pro-jects/variable-power-supply-for-up-to-3a-current)). I want to design a variable power supply for up to 80 volts. Can someone help me?

Victor

□ Please clarify how the combination of 270 ohms and 4.k can output specific voltages like 5V, 6V, 8V, and 9V when rotating the pot? I think it is possible only if we use fixed resistors for such discrete voltages.

Anirvan Kule

EFY. We will try to publish a DIY article on variable supply for up to 80 volts in a coming issue. Based on the datasheet of LM317, its variable output can be controlled as per following relationship:

$$VO = VREF (1 + R2/R1) + (IADJ \times R2)$$

Here, IADJ is typically 50µA and negligible in most applications.

$$VREF = 1.25V$$

$$R2 = VR1 + R4 (4.7K + 100E)$$

$$R1 = R3 = 270E$$

• • •

FACE RECOGNISING ROBOT

This is regarding the DIY article 'Designing IoT Face Recognition AI Robot' published in September issue (also available at <https://www.electronicsforu.com/electronics-projects/design-iot-face-recognition-ai-robot>). Can the circuit diagram of this project be uploaded? I am finding it difficult to figure out connections based on the instructions and pictures published in the article.

Ehab Pilloor

The author Ashwin Kumar Sinha replies:

There is not much to be shown as regards the connections. You only need to put the camera in the Raspberry PI camera port and connect the OLED display as described in the arti-

cle. The connections can be rechecked at <https://www.electronicsforu.com/electronics-projects/design-iot-face-recognition-ai-robot>

• • •

POWER SUPPLY WITH VA METER

As regards the 'Power Supply With Digital Volt-Amp Meter' DIY article published in June issue, I want to know why the feedback resistor and variable resistor have been chosen with so high values when a 270-ohm/330-ohm feedback resistor along with a 4.7k pot would have been enough for getting up to 15V output?

Anirvan Kule

EFY. The variable output can be controlled using the relationship:

$$VO = VREF (1 + R2/R1) + (IADJ \times R2)$$

Here, IADJ is typically 50µA and negligible in most cases.

$$VREF = 1.25V$$

$$R2 = 22K$$

$$R1 = 680E$$

EFY. You can quickly do such calculations for your designs using the calculator available at <https://www.electronicsforu.com/special/inverting-op-amp-voltage-gain-calculator>

• • •

ERRATA

On page 21, in September 2022 issue, under heading Major Limitations, it is mentioned that "power consumption of human brain is extremely low at around 20 to 35 volts," while it should be watts instead of volts.

Sachin Goel

EFY. We thank Sachin for detecting this error and sharing with everyone!

Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: onedit@efy.in)